

BOGOTOL, Russian Federation

WMO: 295530

Lat: 56.2291N

Lon: 89.5535E

Elev: 954

StdP: 14.20

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.0	-27.1	-38.4	0.6	-32.7	-33.0	0.9	-26.8	20.8	17.8	18.3	13.1	2.1	50	0.593

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.8	82.4	66.2	79.3	64.6	76.3	63.2	68.9	77.9	67.0	75.7	65.1	73.8	5.2	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.8	98.6	73.2	63.6	91.3	71.7	61.7	85.1	69.9	33.7	78.6	32.2	75.6	30.6	73.8	76.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.4	14.2	12.4		-36.2	87.3	9.0	2.6	-42.6	89.2	-47.9	90.7	-52.9	92.2	-59.4	94.1
				-36.3	71.7	8.9	2.7	-42.8	73.7	-48.0	75.3	-53.0	76.8	-59.4	78.8
				DB											
				WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	33.9	1.3	6.9	20.9	36.2	48.8	61.8	64.1	59.5	48.3	34.5	17.2	5.9
	DBStd	24.59	15.66	13.73	11.09	10.07	10.14	7.06	5.68	6.42	8.16	10.08	13.62	14.72
	HDD50	7180	1509	1207	904	425	151	8	0	7	129	488	985	1367
	HDD65	11476	1974	1627	1369	865	511	137	84	193	503	946	1435	1832
	CDD50	1308	0	0	0	9	114	361	438	301	78	7	0	0
	CDD65	128	0	0	0	0	8	40	58	21	1	0	0	0
	CDH74	1154	0	0	0	2	99	424	428	175	26	0	0	0
	CDH80	203	0	0	0	0	22	80	67	31	3	0	0	0
Wind	WSAvg	5.3	5.7	5.6	6.3	6.0	5.6	4.3	3.6	3.8	4.7	5.8	6.1	6.2
Precipitation	PrecAvg	20.5	0.9	0.8	0.9	1.2	1.7	2.6	2.3	2.8	2.0	2.1	1.8	1.5
	PrecMax	27.1	1.5	1.6	2.4	2.2	4.4	4.8	4.1	6.5	4.0	3.4	2.7	2.7
	PrecMin	15.9	0.2	0.2	0.1	0.2	0.6	0.5	0.7	1.1	0.2	0.9	1.0	0.4
	PrecStd	2.7	0.4	0.4	0.5	0.5	0.9	1.0	1.0	1.1	0.9	0.6	0.6	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.2	35.3	47.3	67.1	83.0	86.2	85.0	84.0	77.8	63.0	46.8	33.9
		MCWB	29.4	30.5	38.7	50.2	62.7	66.9	69.0	69.3	61.4	49.9	42.1	31.1
	2%	DB	28.4	31.5	41.4	60.9	76.1	82.2	81.9	78.6	70.3	57.4	41.2	30.5
		MCWB	25.5	28.3	34.5	47.4	57.1	64.5	67.8	66.0	57.9	47.7	37.5	28.0
	5%	DB	25.1	28.7	38.4	56.5	71.2	78.9	78.8	74.9	65.9	52.8	36.9	27.7
		MCWB	22.9	25.9	32.9	45.2	54.5	63.2	66.3	63.4	55.1	44.9	33.6	25.8
	10%	DB	21.7	25.3	35.4	51.8	66.3	75.6	75.8	71.7	61.5	48.5	33.3	23.9
		MCWB	19.7	22.5	30.7	42.3	52.6	61.7	64.7	61.7	53.0	42.7	30.7	21.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	29.9	31.2	38.9	51.6	63.3	69.6	71.9	71.2	62.3	52.0	42.7	31.4
		MCDB	32.3	34.3	45.8	63.8	82.2	81.8	81.4	79.6	75.9	60.1	47.1	33.0
	2%	WB	25.9	28.7	35.7	48.8	58.8	66.9	69.4	67.9	58.9	48.3	37.7	28.3
		MCDB	28.2	31.2	40.6	58.9	72.1	77.7	78.3	75.4	68.5	56.3	41.1	30.4
	5%	WB	23.0	26.0	33.1	45.9	56.1	65.0	67.8	65.3	56.5	45.6	33.8	25.8
		MCDB	25.1	28.7	37.5	55.8	67.9	76.0	76.3	72.8	63.5	51.9	36.4	27.8
	10%	WB	19.7	22.6	31.0	42.5	53.6	63.2	66.0	63.0	54.2	42.8	31.0	22.1
		MCDB	21.8	25.3	35.4	50.9	65.2	72.7	73.8	69.9	60.3	48.5	33.2	23.9
Mean Daily Temperature Range	5% DB	MDBR	11.5	12.8	14.4	16.0	18.1	18.5	16.8	16.8	15.6	11.2	10.3	11.4
		MCDBR	11.3	11.9	15.8	22.9	25.3	25.6	21.6	22.4	23.6	18.3	10.4	12.0
		MCWBR	10.1	10.5	11.5	13.3	12.8	11.5	9.9	11.4	12.6	11.0	7.9	11.2
	5% WB	MCDBR	11.4	11.8	14.5	21.5	22.4	21.5	18.8	19.5	20.9	17.4	10.3	11.8
		MCWBR	10.4	10.7	11.3	13.3	12.5	10.7	9.8	10.9	12.1	10.8	8.2	11.1
Clear-Sky Solar Irradiance	taub	0.284	0.317	0.365	0.394	0.405	0.403	0.427	0.406	0.369	0.348	0.314	0.284	
	taud	2.001	1.980	1.923	2.082	2.202	2.297	2.229	2.267	2.318	2.220	2.028	1.945	
	Ebn at Noon	180	224	243	258	264	266	256	254	246	215	164	141	
	Edh at Noon	21	32	44	44	42	38	40	36	30	25	20	17	
All-Sky Solar Radiation	RadAvg	245	551	979	1367	1568	1859	1682	1343	875	450	220	146	
	RadStd	30	49	75	100	174	172	116	109	111	42	31	20	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (14 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page